

Quadratic Functions – Solutions

LET – Mathematics and Statistics

1. Solve by factorising:

(a) $x^2 + 8x + 15 = 0$

Find two numbers that multiply to 15 and add to 8 (5 and 3).

$$(x + 5)(x + 3) = 0$$

$$x + 5 = 0 \quad \text{or} \quad x + 3 = 0$$

$$x = -5 \quad \text{or} \quad x = -3$$

(b) $x^2 + 2x - 15 = 0$

Find two numbers that multiply to -15 and add to 2 (5 and -3).

$$(x + 5)(x - 3) = 0$$

$$x = -5 \quad \text{or} \quad x = 3$$

(c) $x^2 - 5x - 24 = 0$

Find two numbers that multiply to -24 and add to -5 (-8 and 3).

$$(x - 8)(x + 3) = 0$$

$$x = 8 \quad \text{or} \quad x = -3$$

(d) $x^2 + 6x = 0$

Factor out the common term, x .

$$x(x + 6) = 0$$

$$x = 0 \quad \text{or} \quad x = -6$$

(e) $x^2 - 10x + 9 = 0$

Find two numbers that multiply to 9 and add to -10 (-1 and -9).

$$(x - 1)(x - 9) = 0$$

$$x = 1 \quad \text{or} \quad x = 9$$

(f) $x^2 + x = 30$

First, rearrange to equal zero.

$$x^2 + x - 30 = 0$$

$$(x + 6)(x - 5) = 0$$

$$x = -6 \quad \text{or} \quad x = 5$$

2. Solve by factorising (harder):

(a) $3x^2 + 7x + 4 = 0$

Find two numbers that multiply to $3 \times 4 = 12$ and add to 7 (3 and 4). Split the middle term.

$$3x^2 + 3x + 4x + 4 = 0$$

$$3x(x + 1) + 4(x + 1) = 0$$

$$(3x + 4)(x + 1) = 0$$

$$x = -4/3 \quad \text{or} \quad x = -1$$

(b) $5x^2 + 12x + 4 = 0$

Find two numbers that multiply to $5 \times 4 = 20$ and add to 12 (10 and 2).

$$5x^2 + 10x + 2x + 4 = 0$$

$$5x(x + 2) + 2(x + 2) = 0$$

$$(5x + 2)(x + 2) = 0$$

$$x = -2/5 \quad \text{or} \quad x = -2$$

(c) $6x^2 + 7x = 5$

Rearrange to $6x^2 + 7x - 5 = 0$. Find numbers that multiply to $6 \times -5 = -30$ and add to 7 (10 and -3).

$$6x^2 + 10x - 3x - 5 = 0$$

$$2x(3x + 5) - 1(3x + 5) = 0$$

$$(2x - 1)(3x + 5) = 0$$

$$x = 1/2 \quad \text{or} \quad x = -5/3$$

3. Solve using the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$:

(a) $x^2 + 5x + 2 = 0$ ($a = 1, b = 5, c = 2$)

$$x = \frac{-5 \pm \sqrt{5^2 - 4(1)(2)}}{2(1)} = \frac{-5 \pm \sqrt{25 - 8}}{2} = \frac{-5 \pm \sqrt{17}}{2}$$

(b) $x^2 + 7x - 1 = 0$ ($a = 1, b = 7, c = -1$)

$$x = \frac{-7 \pm \sqrt{7^2 - 4(1)(-1)}}{2(1)} = \frac{-7 \pm \sqrt{49 + 4}}{2} = \frac{-7 \pm \sqrt{53}}{2}$$

(c) $3 + 5x - x^2 = 0$ (or $-x^2 + 5x + 3 = 0$, so $a = -1, b = 5, c = 3$)

$$x = \frac{-5 \pm \sqrt{5^2 - 4(-1)(3)}}{2(-1)} = \frac{-5 \pm \sqrt{25 + 12}}{-2} = \frac{5 \mp \sqrt{37}}{2}$$

(d) $4x^2 - x = 1$ (or $4x^2 - x - 1 = 0$, so $a = 4, b = -1, c = -1$)

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(4)(-1)}}{2(4)} = \frac{1 \pm \sqrt{1 + 16}}{8} = \frac{1 \pm \sqrt{17}}{8}$$

4. Analyse the quadratic functions:

(a) For $y = x^2 + 9x + 14$:

i **Y-intercept (set $x=0$):** $y = 14$. Point is $(0, 14)$.

ii **X-intercepts (set $y=0$):** $x^2 + 9x + 14 = 0 \implies (x + 7)(x + 2) = 0$. Roots at $x = -7, x = -2$.

iii **Range for $f(x) > 0$:** Parabola is U-shaped, so positive outside the roots. $x < -7$ or $x > -2$.

(b) For $y = x^2 - 3x + 2$:

i **Y-intercept (set $x=0$):** $y = 2$. Point is $(0, 2)$.

ii **X-intercepts (set $y=0$):** $x^2 - 3x + 2 = 0 \implies (x-1)(x-2) = 0$. Roots at $x = 1, x = 2$.

iii **Range for $f(x) > 0$:** U-shaped, so positive outside the roots. $x < 1$ or $x > 2$.

(c) For $y = x^2 - 8x + 16$:

i **Y-intercept (set $x=0$):** $y = 16$. Point is $(0, 16)$.

ii **X-intercepts (set $y=0$):** $x^2 - 8x + 16 = 0 \implies (x-4)^2 = 0$. One repeated root at $x = 4$.

iii **Range for $f(x) > 0$:** U-shaped, vertex on axis. Positive for all x except $x = 4$.

(d) For $y = 10 + 7x - x^2$:

i **Y-intercept (set $x=0$):** $y = 10$. Point is $(0, 10)$.

ii **X-intercepts (set $y=0$):** $-x^2 + 7x + 10 = 0$. Roots are $x = \frac{7 \pm \sqrt{89}}{2}$.

iii **Range for $f(x) > 0$:** Parabola is n-shaped, so positive between the roots. $\frac{7-\sqrt{89}}{2} < x < \frac{7+\sqrt{89}}{2}$.

(e) For $y = x^2 + 6x - 2$:

i **Y-intercept (set $x=0$):** $y = -2$. Point is $(0, -2)$.

ii **X-intercepts (set $y=0$):** $x^2 + 6x - 2 = 0$. Roots are $x = -3 \pm \sqrt{11}$.

iii **Range for $f(x) > 0$:** U-shaped, so positive outside the roots. $x < -3 - \sqrt{11}$ or $x > -3 + \sqrt{11}$.

(f) For $y = 10 - 3x - 2x^2$:

i **Y-intercept (set $x=0$):** $y = 10$. Point is $(0, 10)$.

ii **X-intercepts (set $y=0$):** $-2x^2 - 3x + 10 = 0$. Roots are $x = \frac{-3 \pm \sqrt{89}}{4}$.

iii **Range for $f(x) > 0$:** n-shaped, so positive between the roots. $\frac{-3-\sqrt{89}}{4} < x < \frac{-3+\sqrt{89}}{4}$.